Case Study 2: ATM Transaction Sequence Simulation

Assignment Case No. 13 | Level: Beginner

01 PROBLEM STATEMENT

Process a sequence of deposits and withdrawals, maintain a simulated balance, enforce withdrawal limits, reject invalid transactions, and produce a transaction summary.

Problem interpretation: A simple ATM simulation must update an account balance for deposits and withdrawals, reject invalid transactions, and prevent withdrawals above the available balance or a per-transaction limit.

02 OBJECTIVES

Process a list of transactions.

Update the balance correctly after each accepted transaction.

Reject invalid amounts and withdrawals that break the rules.

Summarize accepted and rejected transactions.

03 ASSUMPTIONS

The opening balance is non-negative.

Amounts must be positive.

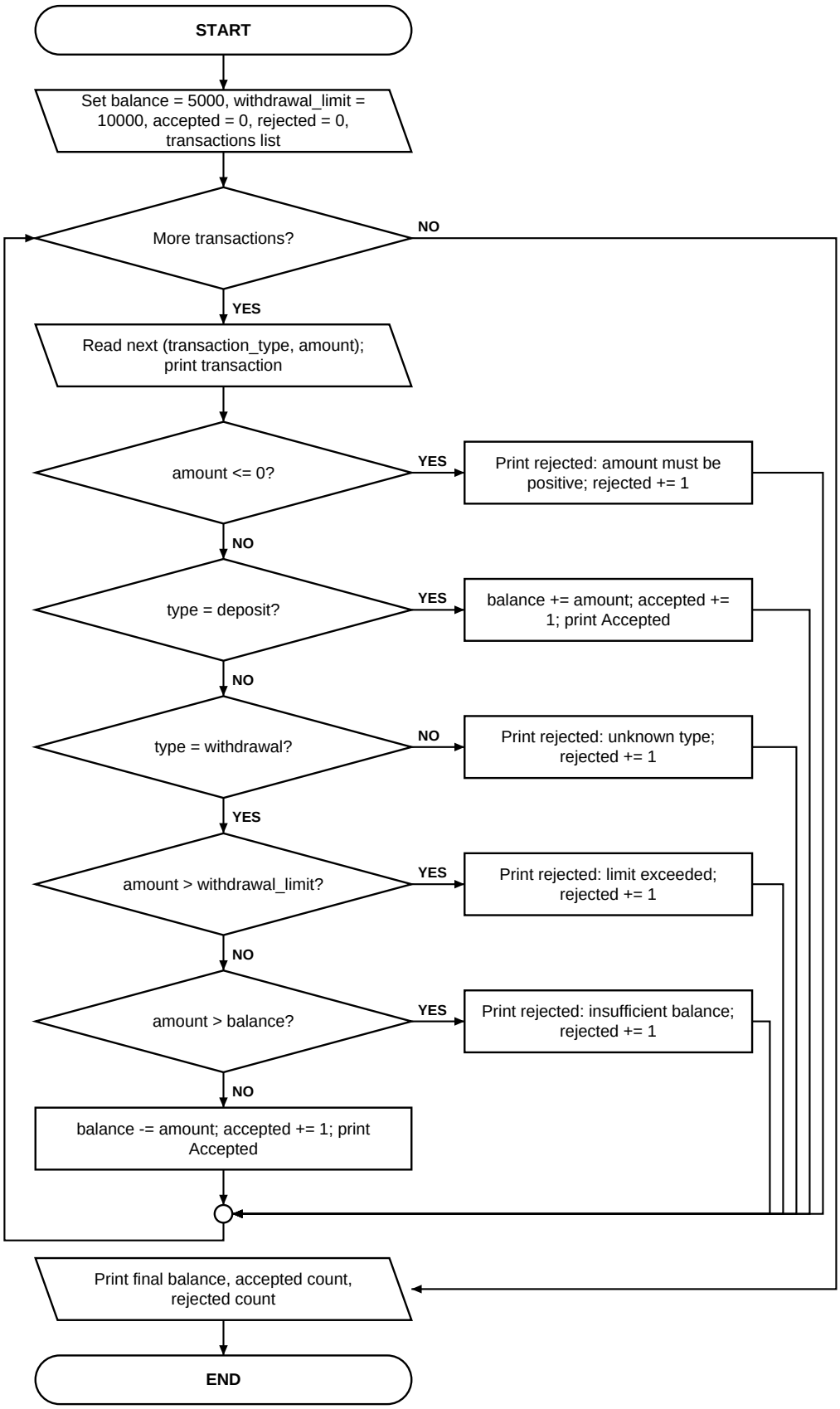
The maximum withdrawal per transaction is ₹10,000.

A withdrawal cannot exceed the available balance.

04 ALGORITHM

1. Start with an opening balance and a list of transactions.
2. Set accepted and rejected counters to zero.
3. For each transaction, validate the amount.
4. For a deposit, add the amount to the balance.
5. For a withdrawal, check the limit and available balance before subtracting.
6. Reject unknown transaction types.
7. Display each transaction result and the final summary.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    balance = 5000; withdrawal_limit = 10000
    accepted = 0; rejected = 0
    FOR each (transaction_type, amount) IN transactions
        PRINT transaction
        IF amount <= 0 THEN
            PRINT "Rejected - amount must be positive"; rejected = rejected + 1
        ELSE IF transaction_type = "deposit" THEN
            balance = balance + amount; accepted = accepted + 1
        ELSE IF transaction_type = "withdrawal" THEN
            IF amount > withdrawal_limit THEN
                PRINT "Rejected - withdrawal limit exceeded"; rejected = rejected + 1
            ELSE IF amount > balance THEN
                PRINT "Rejected - insufficient balance"; rejected = rejected + 1
            ELSE
                balance = balance - amount; accepted = accepted + 1
            END IF
        ELSE
            PRINT "Rejected - unknown transaction type"; rejected = rejected + 1
        END IF
    END FOR
    PRINT balance, accepted, rejected
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 02_ATM_Transaction_Sequence_Simulation_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
deposit ₹2000 from ₹5000	Positive amount, valid type	Accepted; balance becomes ₹7000
withdrawal ₹1500 from ₹7000	Within limit and balance	Accepted; balance becomes ₹5500
withdrawal ₹7000 from ₹5500	Exceeds available balance	Rejected: insufficient balance; balance unchanged
withdrawal ₹10500	Above the 10,000 limit	Rejected: withdrawal limit exceeded
deposit ₹-200	Amount not positive	Rejected: amount must be positive
transfer ₹500	Unknown transaction type	Rejected: unknown transaction type

09 SAMPLE OUTPUT PREVIEW

```
Transaction: deposit ₹2000
Result: Accepted
Transaction: withdrawal ₹1500
Result: Accepted
Transaction: withdrawal ₹7000
Result: Rejected - insufficient balance
Transaction: deposit ₹-200
Result: Rejected - amount must be positive
Transaction: transfer ₹500
Result: Rejected - unknown transaction type

--- Summary ---
Final balance: ₹ 5500
Accepted transactions: 2
Rejected transactions: 3
```

10 CONCLUSION

The simulation demonstrates lists, tuples, loops, conditional statements, and validation while maintaining a correct running balance.